

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

Stockholm

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PROFILE

Experimental physicist (PhD candidate, Stockholm University) working on a large accelerator experiment at GSI/FAIR. Detector work, particle identification, Monte Carlo simulation, statistical analysis, machine-learning classification, and automated analysis pipelines on HPC. Earlier industry experience in automotive radar algorithms and electromagnetic engineering.

Completing my PhD in 2027 and open to industry R&D and data roles. Sectors of interest: Sensing & defence systems • Measurement technology & metrology • Medical technology & radiation • Autonomous systems • Applied R&D • Quantitative analysis

FROM THE LAB TO INDUSTRY

Detectors & sensor systems

Specify, integrate and debug multi-sensor systems; reason from raw sensor behaviour to system performance.

Signal from noise: statistics & uncertainty

Robust inference, detection and estimation with honest uncertainty budgets for decisions that depend on noisy data.

AI/ML & AI-assisted workflows

Apply ML to physical and sensor data with domain understanding, and use modern AI tools to ship analysis software faster.

Measurement, calibration & instrumentation

Design calibration strategies, characterise instruments and keep measurements traceable and reproducible.

Simulation & Monte Carlo

Build digital twins and synthetic data, run sensitivity and feasibility studies before hardware exists.

Large-scale data & HPC

Engineer data pipelines that scale from a notebook to a compute cluster, with reproducible, reviewable code.

EXPERIENCE

PhD Candidate (Doktorand)

Stockholm University • PANDA@HADES, GSI/FAIR

Aug 2021 – present

Stockholm, Sweden & Darmstadt, Germany

Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR.

- Main analysis responsibility in PANDA@HADES for the $\Lambda\Lambda$ channel in proton–proton collisions at $\sqrt{s} = 3.46$ GeV.
- Took detector shifts during the five-week February 2022 proton beam time and a 2025 pion-beam run; worked with drift-chamber online quality assurance.
- Developed a relative time-of-flight particle-identification method that compensates start-detector timing problems.
- Trained multiclass machine-learning classifiers (ROOT TMVA) to separate $\Lambda\Lambda$ signal from background.
- Built an automated analysis pipeline with systematic variations on the GSI batch farm (Slurm / HPC), developed with large language models.
- Wrote an event generator and detector-response folding for CBM feasibility studies of the $\Lambda\Lambda$ spectrum and the ΞN cusps; contributed to the QCD at FAIR white paper.

C++ • ROOT • Python • TMVA • Slurm • PLUTO • GEANT • KinFit • Git

Electric Propulsion Engineer

KTH Hyperloop team

Sep 2018 – 2019

Stockholm, Sweden

High-energy propulsion unit for the SpaceX Hyperloop Pod Competition 2019.

- Helped design a high-energy propulsion unit, a double-sided linear induction motor for the pod; electromagnetic finite-element simulations in COMSOL and ANSYS Maxwell, with MATLAB.

COMSOL • ANSYS Maxwell • MATLAB

Associate Software Engineer

Robert Bosch Engineering and Business Solutions (RBEI)

Aug 2017 – Jul 2018

Bengaluru, India

- Worked on signal-processing algorithms for automotive radar sensors used in driver-assistance systems, in an industrial software-engineering environment.

EARLIER PROJECTS & INTERNSHIPS

Intern, Network Engineering Facility, AT&T, Bengaluru. VPN order-management processes in a network-engineering organisation.

Jan–Jul 2017

Hybrid levitation for Hyperloop: engineering project, BMS College of Engineering, Bengaluru. Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept.

Aug 2016 – Jun 2017

Intern, Quality Assurance division, ISRO, Laboratory for Electro-Optics Systems (LEOS), Bengaluru. Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions.

Jul 2015

EDUCATION

PhD , Experimental Nuclear & Hadron Physics · <i>Stockholm University</i>	2021–2027 (expected)
Licentiate of Philosophy (Fil.lic.) , Physics · <i>Stockholm University</i>	2021–2024
Thesis: “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions”	
MSc , Electromagnetics, Fusion and Space Engineering · <i>KTH Royal Institute of Technology</i>	2018–2020
BE , Electronics and Communication Engineering · <i>BMS College of Engineering</i>	2013–2017

CORE SKILLS

Data analysis, ML & AI	Statistical inference (maximum-likelihood & template fits, least squares, sideband methods) · Uncertainty quantification (systematic variations, correlated averages) · Kinematic (constrained) fitting (vertex and multi-constraint fits) · Machine-learning classification (multiclass classifiers in ROOT TMVA) · AI-assisted development (large language models for building and automating pipelines)
Programming & software	Python (NumPy, SciPy, pandas, matplotlib, uproot, iminuit, Jupyter) · C++ & ROOT (analysis frameworks, PyROOT, RooFit) · Git / GitLab
Simulation & modelling	Monte Carlo event generation (PLUTO, own generator with final-state interactions) · Detector simulation (GEANT3-based HGeant, CbmRoot) · Finite-element electromagnetics (COMSOL, ANSYS Maxwell)
Detectors & instrumentation	Tracking & timing detectors (drift chambers, straw tubes, RPC, LGAD start detector, time of flight) · Gamma spectroscopy (HPGe detectors, scintillators, neutron activation) · Beam-time operations & online data-quality monitoring · Radiation safety & radioactive-source handling · Automotive radar signal processing
Computing & HPC	HPC batch computing (Slurm on the GSI Virgo cluster) · Containers & software distribution (Singularity/Apptainer, CVMFS) · Large datasets (~155 TB raw data, Lustre, ROOT trees, HDF5)

SELECTED PUBLICATIONS & TALKS

G. Appagere (PANDA@HADES), “Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores”, <i>JPS Conference Proceedings, 15th International Conference on Hypernuclear and Strange Particle Physics (HYP2025)</i> , Tokyo. IN PRESS	2026
J. G. Messchendorp, F. Nerling, ..., G. Appagere , et al., “Hadron Physics Opportunities at FAIR”, <i>arXiv:2512.15986 [hep-ex]</i> . PREPRINT	2025
9 conference talks , including NSTAR 2026 (Seville), 15th Nordic Meeting on Nuclear Physics (Visby, 2026), DPG Spring Meeting 2026 (Erlangen) and HYP2025 (Tokyo); plus 6 presentations at internal HADES and CBM collaboration meetings.	
	2022–2026

LABORATORY

Lab responsible , Nuclear physics student laboratory, Fysikum. Handled the radioactive sources for the laboratory through these five years. Responsible for HPGe detectors, scintillators and the source inventory; lab rules and safety, a source booking and record system, and procurement of relevant laboratory components, including detectors and NIM units.	2021–2025
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LEADERSHIP & RESPONSIBILITY

Chair (2025–2026), Vice-Chair (2023–2025), Fysikum PhD Council, Stockholm University.	2023–2026
Vice-Chair , Central Doctoral Student Council (CDR), Stockholm University.	2025
Vice-Chair , Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University.	2023–2024
Doctoral-student representative , Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment).	2023–2026
University teaching , Stockholm University. Laboratory courses and classroom teaching, including lectures, tutorials and problem solving, in 6 physics courses; co-supervision of a Bachelor’s thesis in Medical Radiation Physics.	2021–2025

AWARDS & LANGUAGES

Awards	SAE E-Baja 2017: 3rd place in India, design review · IEEE Darrel Chong Student Activity Award (2016) · SAE Aero Design West 2015: 13th worldwide (team captain)
Grants	4 grants and scholarships for research travel and experiments (Stockholm University, 2022–2026)
Languages	English: Fluent · Swedish: Basic (A2)